

Gustavo Costa

NLP researcher with a psychotherapy and philosophy background

📍 London, UK | ✉️ gustavo@arcos.org.br | ☎️ +55 61 98133-4405 | [in](https://www.linkedin.com/in/gustavoarcos) [gustavoarcos](https://www.linkedin.com/in/gustavoarcos)

🌀 [noah-art3mis](https://noah-art3mis.com) | 🌐 simulacro.tech

Interests

Computational social science: AI-generated misinformation, scientific integrity, fact-checking, mechanistic interpretability for hypothesis generation

Philosophy: Philosophy of science, science and technology studies, neopragmatism (Popper, Haraway, Latour, Rorty)

Education

London Interdisciplinary School, MSc in Interdisciplinary Problems and Methods 2024 – 2025
– London, UK

- Hackathon Winner: AI Safety Hackathon (Holistic AI & UCL AI Society) – Stereotype detection in large language models using sparse autoencoders
- Student representative on the Equality, Diversity and Inclusion Committee
- Grade: Distinction

Faculdade Inspirar, Specialist in Lacanian Psychoanalysis – Brasilia, Brazil 2019 – 2022

- Grade: 9.1/10

University of Brasilia, Bachelor in Psychology – Brasilia, Brazil 2014 – 2019

- Student representative in the Academic Centre
- Grade: 4.5/5.0

Professional Experience

Lead Engineer, LLM Systems & Research Infrastructure, CAPTA – Remote (Mexico City, Mexico) Jan 2026 – present

Designed LLM-based systems and internal AI tools for political focus group research in Mexican Spanish. Co-developed with humanist researchers, integrating into existing workflows rather than replacing them. Reported directly to CEO.

- Built Adapta: 7-stage data pipeline and LLM fine-tuning infrastructure (QLoRA) with 40+ evaluation metrics. 100+ training runs, 165 GPU hours, 22 empirical findings.
- Built Nopalero: automated participant screening system (48 validation checks, fraud detection, OCR, socio-economic classification) replacing hours of manual data entry per project.
- Built Molcajete: audio-to-insight pipeline with speaker diarization, transcription, theme classification, and integrated reporting. 1,300+ hours of audio across 10 Mexican states; processes a full project in 30–60 minutes.

Psychotherapist, Espaco Psique Psychology Clinic – Brasilia, Brazil 2020 – 2024

- Provided psychoanalytic counselling and produced psychological reports.
- Published essays on philosophy of psychology at gustavocosta.psc.br [↗](#).

Research Experience

Master's Dissertation: 'What is AI slop?'

Mixed-methods research on AI slop, developing a four-dimensional taxonomy (themes, metaphors, qualities, types) and comparing linguistic patterns of AI-generated and human-written news.

- Integrated qualitative and computational methods, drawing on computer science, philosophy, psychology, and media studies.
- Compared linguistic patterns in AI-generated vs. human-written news using topic modelling, sentiment analysis, and LLM-based classification.
- Curated a new web-scraped dataset and disseminated findings through essays on simulacro.co.uk.

Technical Projects

Judex-mini [↗](#) Jan 2025
Python library for scraping Brazilian Supreme Court case data.

Intersect [↗](#) Jan 2025
Job board that uses NLP (ML, embeddings, reranking) to identify more relevant roles. Deployed as streamlit app.

LexGPT [↗](#) 2023 – 2024
AI services for lawyers. RAG app with flask backend and vanilla JavaScript frontend.

Qualitative Projects

simulacro.co.uk [↗](#) 2024 – present
Essays on data science, AI ethics and philosophy of science.

Fora Freud [↗](#) 2021 – 2024
Essays on philosophy and psychoanalysis, in Portuguese.

@simulacro.psi [↗](#) 2021 – 2024
Instagram page that features a daily handpicked philosophy quote (1.5k followers).

Publications and Talks

Xenofeminism Jan 2025
(Thinking Out Loud, London Interdisciplinary School)

The midnight sun: Popper and rationality without foundations Jan 2023
(Psychoanalysis and Science Research Group, APOLa SP)

Skills

Programming: Python, git, HTML/CSS/JavaScript/TypeScript

Data Science: Data cleaning, data mining, data visualization (Pandas, NumPy, Matplotlib, Seaborn)

Natural Language Processing: Text classification, topic modelling, sentiment analysis, embeddings (spaCy, NLTK, Gensim, Flair, BERTopic, sentence-transformers)

Machine Learning: Model training, evaluation, and interpretability (transformers, scikit-learn, PyTorch, SHAP, LIME, Sparse Autoencoders)

AI Engineering: LLM fine-tuning (QLoRA, Unsloth), inference (vLLM), evaluation (W&B/Weave), cloud GPU (Modal), RAG, context engineering

Languages: Portuguese (Native), English (Fluent), Spanish (Professional)

Certificates

- HarvardX Introduction to Artificial Intelligence
- Introduction to Contemporary Science Studies: Weaving Knowledge Practices (APPH)
- Living and Dying in the Chthulucene: Donna Haraway's Ethics of Care (APPH)
- Psychoanalysis and Gender Dissidence: Transphobia, Power, and Resistance